

GOD ONLY KNOWS ARISTOTLE ON MIND AND GOD

How can you tell if someone is a philosopher? There are a number of potential clues. Does this person like to argue? Do they have a habit of staring off silently into the middle distance? Do their grooming and hygiene leave something to be desired? Chances are you're in luck: you've found yourself a philosopher. But an almost sure-fire test is this. Tell them that you yourself are a philosopher. If you are dealing with a normal person they'll say something like, "Gosh. So, um, what do you do all day?" But if you're talking to a philosopher, they will ask you what area of philosophy you work in. Professionals know that philosophy has many branches, and that no one, these days, is a "philosopher" in the sense Aristotle was: a pursuer of all the knowledge that mankind can hope to possess.

In the English-speaking world, the philosophers you meet at parties will often say, when asked what area of philosophy they do, that they are philosophers of mind. Not that philosophers of mind go to many parties. They are too busy debating issues like the nature of consciousness, the relation of our minds to our bodies, and the philosophical relevance of empirical research in brain science and psychology. This is cutting-edge stuff, but like most, if not all, areas of philosophy, it has ancient roots. Aristotle was arguably the inventor of the philosophy of mind, just as he was the inventor of logic, biology, and numerous other disciplines. Of course, when I was talking about his work *On the Soul*, I did say that it is a mistake to confuse soul, as Aristotle understands it, with the topic considered by today's philosophers of mind. Aristotle's soul is rather the explanation for all our vital functions, including nutrition, motion, and sensation as well as thought. So when Aristotle talks about the relation between soul and body, this is not the same as when Descartes, or a modern-day philosopher, talks of the mind-body relationship.

Yet Aristotle does provide an account of thinking, in the third book of *On the Soul*. Here he takes up the topic of mind, not in the sense of all our experiences, or of consciousness—what philosophers now call "mental" phenomena—but rather mind in the sense of intellect. The Greek word for "intellect" is *nous*. It refers here specifically to the capacity by which we achieve knowledge. Aristotle considers this

to be an exalted, even divine, capacity which distinguishes humans from lower animals. When he concludes his *Ethics* by suggesting that a life of contemplation is the best life for humans, he means that our greatest happiness would be achieved through the exercise of intellect. And yet, Aristotle sees close ties between intellect and the lower functions of sensation and imagination. We already saw that, at the end of the *Posterior Analytics*, he claims that sensation is the basis for the first principles of demonstrative knowledge—and he calls the state of grasping those principles by the same word, *nous*, or intellect.

Here in *On the Soul*, meanwhile, Aristotle draws a close parallel between intellect and sensation. As we've seen, he understands sensation as the actualization of a potentiality. If your eyes function properly, then even when you are not seeing anything, you are potentially seeing red; you have the capacity to see red. When you are confronted with a red apple, that capacity is realized. Your sight becomes not just potentially red, but actually red, which is just to say that you see red. The same kind of story can be told for the intellect. Your intellect is a capacity you have, thanks to your possession of a human soul. When it is actualized, what you do is think. But what Aristotle seems to mean by "thinking" here is something rather special. He doesn't mean thinking about what you'll have for dinner, say, or trying to remember where you left your keys. He means having a grasp of essences or forms out in the world. When you think in this way about giraffes, you actually grasp the nature of giraffe, just as vision actually grasps the redness of an apple. So we might think of intellect as relating to natures the way vision relates to colors.

Despite the analogy between thinking and sensation, there are, of course, also differences. One difference is that there is no falsehood in sensation. Sight can provide you with a *misleading* appearance, like the mirage in the desert that looks like an oasis (or indeed the sound made by a tribute band that sounds like Oasis). But according to Aristotle, sight itself is always true, as is any kind of sensation (428a)—the falsehood would appear at the level of a judgment made on the basis of this misleading impression. Falsehood already appears in a faculty Aristotle calls *phantasia* (often translated, rather inadequately, as "imagination").¹ Imagination is distinct from both sensation and thought, but intimately connected to both. Imagination is always based on what we have perceived with the senses, though we may manipulate what we have sensed (like combining the images of a human and a horse to get a centaur), and thinking cannot occur without imagination (427b, 431a). It is perhaps clear enough that sensation isn't the same thing as imagination, but one might be tempted to identify thinking with imagination. This would be a mistake, says Aristotle, since belief only comes in at the level of thinking. When you imagine, you are, as it were, entertaining a picture of how things might be, but you are not

convinced that things actually are this way. Imagination is “false” only in the sense that it does not represent reality as it is. But merely imagining things to be a certain way does not mean you believe they actually are that way. False belief arises only in thought, through combinations of terms that don’t correspond to reality (430b—we might remember the similar point made by Plato in the *Sophist*, with the example “Theaetetus flies”). Aristotle thinks this marks a dividing-line between us and the non-human animals: they are capable of imagination, but not belief (427a).

Another difference between thinking and sensation is that thinking happens without the use of any bodily organ (429a). It is realized through our souls, but not through any part of our body. It’s no surprise that Aristotle fails to identify the brain as the seat of intellect. For him, the brain is basically just a refrigeration unit, designed to balance the heat stemming from our heart.² It is the heart that serves as the command center; for instance, it is the ultimate seat of sensation. But we don’t use it to think, nor do we use any other part of the body. Aristotle’s argument for this begins from the observation that you can think about anything: everything is a possible object of intellection. And intellect thinks by actually taking on the form of what it thinks about. This means that if it were seated in an organ that were, for instance, hot, it would always be thinking about heat, because it would always be actually hot. But this is clearly not the case. Rather, in its basic state intellect is only *potentially* everything. So it cannot have any bodily organ, given that every organ already has many actual features.

This argument probably won’t be keeping my colleagues who do philosophy of mind up at nights. They would already be puzzled by the idea that thinking is taking on a form. And even granting this to Aristotle, surely the sense in which the intellect takes on a form could be different from the sense in which a bodily organ has that form. Still, the argument does underscore Aristotle’s commitment to an idea that will have a far-reaching historical impact, namely that our minds become identical in form to whatever we think about. He’s also being consistent with what he’s said about sensation. Regarding vision, for instance, he remarked that the fluid in the eye must have no color, since otherwise we would always be seeing that color. Rather, it must be potentially all colors. In the same way, the intellect is potentially everything.

The idea that we think without using a bodily organ gives us a hint as to why Aristotle believes that intellect is something divine. As he says in the *Ethics*, the life of the mind is not so much the best life for us, but a life which is, if anything, too good for mere humans. Really, it is divine. This isn’t just a figure of speech. As we can see if we turn to works other than *On the Soul*, Aristotle in fact portrays God as a pure intellect. To piece together his account, we need to return to his cosmology, as it is laid out in the *Physics* and another treatise called *On the Heavens*. In these writings he

describes the cosmos as spherical in form, made up of many nested spheres. At the edge is the outermost sphere, in which are embedded the so-called “fixed stars”—the ones that, when seen from earth, appear to rise and set every night, and to move together. Other stars can be seen to change position from night to night, wandering against the background of these fixed stars. These are the so-called “planets” (the word comes from the Greek verb meaning “to wander”). The planets are embedded in further spheres nested inside the outermost sphere of fixed stars.

Our lower realm is beneath these heavenly spheres, at the center of the cosmos. It is made up of the four elements, which are mixed together to produce the earth, the seas, the sky, and all the plants, animals, and other substances we see around us. It would seem that the celestial motion is somehow responsible for the mixture of these elements. And the celestial motion can be traced back to the motion of the outermost sphere. For it is the daily rotation of this sphere that accounts for most of what we see in the night sky. The wandering of the planets is due to slight changes in their rotation, introduced by the various motions of the spheres upon which they are seated. But basically the planetary spheres likewise go around once per day, carried around by the outermost sphere.

Thus, if we want to ask what ultimately explains all motion, all generation and destruction, in our cosmos, we need to find the cause of the eternal motion of the outermost sphere. It will be this mover that sets in chain the complex series of celestial rotations that mix up the four elements to yield the environment we live in. When giraffes lope across the savannah, that motion can be traced—very indirectly—back to the single, simple rotation of the outermost sphere. I think I’m not giving anything away if I now say that the cause of this rotation is God, an unmoved mover (*Physics* 260a). But why think that this motion, or any celestial motion, has any cause at all? Why not just say that the motion is a brute fact about the cosmos? This question presents itself with particular force in light of Aristotle’s belief that the world is eternal, a central claim in both the *Physics* and *On the Heavens*. Aristotle is convinced that the world has not only always existed, but has always looked much as it does now, with the same heavenly rotations, the same animal and plant species. Although Aristotle does want to find a place for God in his cosmology, his God is not a creator. So again: why think that the heavens need any explanatory principle?

Aristotle gives several answers here—though perhaps the basic underlying idea is simply that the eternal heavenly motion cannot just happen to be the way it is, but must have a cause. This would bring the heavens within Aristotle’s general conception of nature, where nothing is unexplained and everything has a purpose. But he wants to establish exactly what this cause will be like. So he gives the following

argument. The motion of the heaven is eternal, and thus in a sense infinite. But nothing finite can have enough power to perform an infinite motion (*Physics* 266a–b). Imagine trying to run a car forever on a finite amount of fuel. However efficient the car is, it will run out of fuel eventually. Now, the heavens are finite, since they are bodies of enormous, but nonetheless limited, size. So they cannot have the power to move eternally under their own steam. They need an external mover. Obviously this mover cannot have a body, since if it did it too would be limited and finite. So the external mover “has no magnitude”—that is, it is immaterial (267b).

At the risk of complicating matters, I should say that Aristotle’s argument will apply to each individual heavenly sphere. There is not only the outermost heaven, but also each of the planetary spheres. Actually it’s even worse, since Aristotle, following the astronomical theory produced in Plato’s Academy by men like Eudoxus, believed that each planetary motion must be explained as a combination of several simple rotations. This explains why the planet moves in a wandering path across the sky from night to night. Aristotle seems to have improved upon the mathematicians here, by positing several actual physical spheres for each required planetary motion, with further spheres to counteract those motions to prevent them from being passed on to lower planets (*Metaphysics* 1073a–1074a). Eudoxus and others had simply observed that combining enough circular motions would yield the right mathematical calculation for expressing, say, the motion of Venus, but they postulated no physical mechanism. Aristotle instead assumes that there are real physical spheres, and to account for planetary motions we see we will need several dozen of them, each of which needs its own immaterial mover (1074a). So God is, for Aristotle, simply the greatest and most important of the bodiless celestial movers. As Aristotle says, earlier thinkers had a vague intuition of this truth when they spoke of many gods (1074b). There are indeed numerous divine beings, but they are not the squabbling family of Hesiod and Homer. Aristotle follows the lead of Xenophanes and Plato by scrubbing the “gods” clean of almost all human features. They are simply separate, immaterial substances, and they cause the various heavenly spheres to move.

So far, so good, except that it’s hard to see how immaterial beings can make spheres rotate. Not, presumably, by getting out and pushing. Aristotle does have an answer to this question, which he gives in the twelfth book of his *Metaphysics*. Here Aristotle presents his theory of the divine as an improvement on Plato. He has captured something of what Plato wanted to do with his theory of Forms, by tracing physical phenomena back to immaterial causes. But whereas the Forms, according to Aristotle at least, can’t really explain motion (1070a), Aristotle’s celestial movers will give rise to simple, circular motions—which go on to cause everything else.

Aristotle is no doubt right that it is a mystery how the Form of Large causes things to be large. But his story will only count as an improvement if he can tell us how God moves the outermost heaven, and how the other divine movers move their spheres.

When Aristotle answers this question, he winds up admitting that the divine does, after all, have something in common with humans. For the sole activity of the divine movers turns out to be a distinctively human activity: thinking (1072b). These movers are nothing but separate intellects, and the spheres move in imitation of their eternal thinking by performing eternal circular motions. So Aristotle describes God as a kind of final cause, saying that it moves the outermost heaven the way that the beloved moves the lover (1072a). Just as the pretty girl at the dance makes the boys shuffle shyly towards her without doing anything herself, so God can without moving cause other things to move, out of their aspiration to be like the divine. Here again we have an idea that will inspire centuries of philosophers, and it isn't hard to see why. For Jewish, Christian, and Muslim readers, the notion that God is the final object of love and desire in the universe was, if you'll pardon the pun, highly attractive.

Still, though, has Aristotle really given us a story about why God and the other movers are *thinking*? Why couldn't they be doing something else? Granted, he has argued that they are immaterial. And for Aristotle, the activity that we perform without using any part of our body is thinking. This seems to have been enough to make him leap to his conclusion. God is immaterial and he must be doing something; when we do something immaterially we are thinking; therefore God thinks. This simply assumes, rather than arguing, that there is only one kind of immaterial activity. But there are a few things Aristotle could say in his defense. He might insist that it is needlessly complicated to presuppose further immaterial kinds of activity: if we know of one already, namely thinking, why postulate another one? Moreover, he might argue that thinking simply is immaterial activity, that the two are one and the same. After all, what is it for us to think, apart from possessing form immaterially? So if God is doing something, and is immaterial, it follows more or less from Aristotle's definition of thinking that God is thinking.

Even if we grant this, we'll want to know at least one more thing: what is God thinking about? Aristotle responds that, since we're talking about God, we need to allow him to think about the best possible object. There are some things that it would be better not to think about at all. God is clearly not thinking about where he left his keys. In fact, if God is going to think about the absolutely best thing, there's only one candidate: himself. God will be, as Aristotle famously puts it, "thought thinking about thought," sometimes paraphrased as "thought thinking itself" (1072b).

This phrase is famous, but also a bit disturbing. Are we being presented with a divine navel-gazer, a God who does nothing day after day, for all eternity, but think about himself? This sounds disappointing. After all, if God is nothing but an intellect who thinks himself, then the only thing he can think about is, what? The fact that he's thinking about himself? Sounds not only self-absorbed, but rather pointless.

Some interpreters assume that, for Aristotle, God thinks about everything, but indirectly. Since he is the cause of the first celestial motion which leads to all the other motions, he could grasp all things as implications of his own causal influence. But Aristotle certainly doesn't say this explicitly, and actually I don't think he was too worried about the *content* of divine thought. I would guess that he just meant to say that whatever God thinks about, God must be permanently self-aware. Just as we, at our best moments, consciously reflect on our own knowledge—and what is philosophy, if not such conscious self-reflection?—so God permanently thinks of Himself as thinking about whatever else it is that he thinks about. The extent of his knowledge regarding other things may be left as an open question.³

At any rate, it's clear that Aristotle seriously intends us to regard this first celestial mover—the so-called “prime mover”—as worthy of respect and perhaps even something like worship. I'm not saying that he wants us to go to the temple of Zeus and sacrifice animals to the prime mover. But he does end his discussion of God in the *Metaphysics* with an unusually rhetorical passage, identifying the prime mover as the ultimate source of order in our cosmos (1075b–1076a). It not only causes the motion of the outermost heaven, but also somehow coordinates all the other motions. Quoting a line from Homer, Aristotle finishes with a flourish, saying “the rule of many is not good; let there be one ruler” (1076a). He seems to be hinting that his God is not just perfect, but also providential. Again, authors in future generations will take note, and take advantage, weaving Aristotle's theology together with the teachings of revealed religion.

Aristotle's God is very much a god of the philosophers, in every sense of that phrase. The most appropriate type of worship is no doubt what Aristotle himself does: to contemplate God through reasoned argument. And God is himself a kind of philosopher, but better, a pure intellect who always thinks, always thinks about his thinking, and never has to go to sleep. God thus provides a standard for all other things to imitate. The heavens imitate his thinking by their unending celestial rotations. Even plants and animals imitate the divine through reproduction, which allows them to attain a kind of immortality of their own—their species will live on endlessly, although the individuals die. As for us, we can imitate God by doing philosophy. If this all leads Aristotle's readers to infer that Aristotle is the most godlike man they know, that would probably be okay with him. But there is another

way for us to imitate God. God is the ruler of all things, who brings order and harmony. We can do something similar in our practical affairs. By setting up sound systems of governance in our cities, we humans can imitate the order of the universe—an idea already endorsed by Plato in the *Republic* and *Timaeus*. What systems would produce the best possible order? To find out, we'll need to turn to another Aristotelian treatise, which is in no small part a response to the *Republic*, and which handles its political proposals just as roughly as the *Metaphysics* handles Plato's theory of Forms.